

### Project Executable Files

|                      |                                                |
|----------------------|------------------------------------------------|
| <b>Date</b>          | 5 July 2026                                    |
| <b>Team ID</b>       | xxxxxx                                         |
| <b>Project Name</b>  | OptiCrop – AI-Based Crop Recommendation System |
| <b>Maximum Marks</b> | 3 Marks                                        |

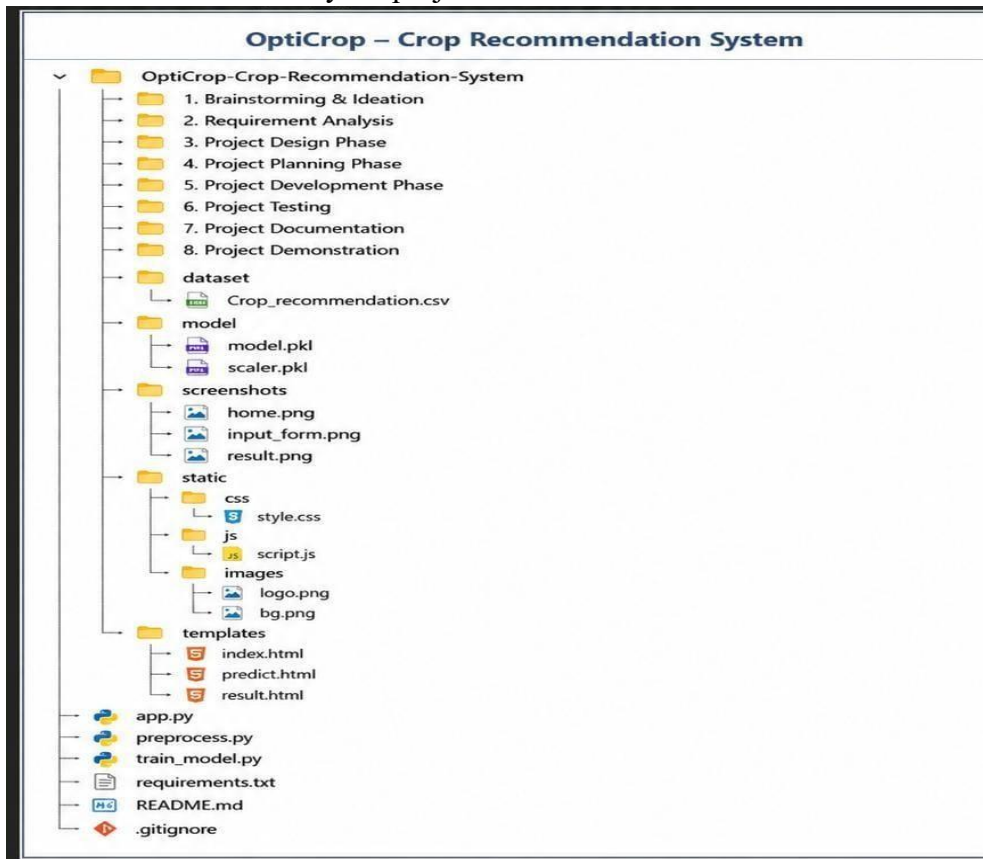
### Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions. **Step 1: Submission Checklist**

| S.No | Item to Submit                                                  | Submitted (Yes / No / NA) |
|------|-----------------------------------------------------------------|---------------------------|
| 1    | Complete source code (all files and folders)                    | Yes                       |
| 2    | README / Setup Guide (instructions to run the project)          | Yes                       |
| 3    | requirements.txt / package.json / dependency file               | Yes                       |
| 4    | Database schema / seed files (if applicable)                    | NA                        |
| 5    | Environment or configuration file (.env.example similar)        | NA                        |
| 6    | Deployed application URL (if hosted)                            | Yes                       |
| 7    | APK / Executable binary (if applicable for mobile/desktop apps) | NA                        |
| 8    | Dockerfile / Containerization config (if applicable)            | NA                        |
| 9    | Test files and test results                                     | Yes                       |
| 10   | Demo video or walkthrough (if required)                         | Yes                       |

## Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:



## Step 3: Deployment / Access Details

| Field                       | Details                                                                                                       |
|-----------------------------|---------------------------------------------------------------------------------------------------------------|
| Hosted / Deployed URL       | <a href="https://opti-crop-ai-1.onrender.com">https://opti-crop-ai-1.onrender.com</a>                         |
| Login Credentials (Demo)    | Not Required                                                                                                  |
| Platform / Hosting Provider | Render                                                                                                        |
| Repository Link             | <a href="https://github.com/farru052000-ops/OPTI_CROP_AI">https://github.com/farru052000-ops/OPTI_CROP_AI</a> |

|                        |                                                                                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Demo Video Link</b> | <a href="https://drive.google.com/file/d/1qusg9HHm0hkd3UIEnRa2E3bqs8l3w8Zf/view?usp=drivesdk">https://drive.google.com/file/d/1qusg9HHm0hkd3UIEnRa2E3bqs8l3w8Zf/view?usp=drivesdk</a> |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Run Instructions – OptiCrop – Crop Recommendation System

- 1



Download or clone the OptiCrop project from the GitHub repository.

```
git clone https://github.com/23ata05317/OPTICROP.git
```
- 2



Open the project folder in Visual Studio Code or any Python IDE.

```
cd OPTICROP
```
- 3



Create a virtual environment:

```
python -m venv myenv
```
- 4



Activate the virtual environment:

```
On Windows      : myenv\Scripts\activate
On Mac / Linux   : source myenv/bin/activate
```
- 5



Install all required dependencies:

```
pip install -r requirements.txt
```
- 6



Ensure the dataset (Crop\_recommendation.csv), trained model (model.pkl), and scaler (scaler.pkl) are placed in their respective folders.
- 7



Run the Flask application:

```
python app.py
```
- 8



Open a web browser and navigate to:

```
http://127.0.0.1:5000/
```
- 9



Enter the required input details and click **Predict** to get the recommended crop.

### Step 5: Known Issues / Limitations

| S.No | Known Issue / Limitation                                                                             | Workaround / Status                                                                                     |
|------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1    | Crop recommendation accuracy depends on the quality and diversity of the training dataset.           | Improve the dataset by adding more real-world agricultural data and retrain the Machine Learning model. |
| 2    | The application does not provide user authentication or store prediction history.                    | User login can be implemented in future versions.                                                       |
| 3    | On Render's free hosting plan, the application may take a few seconds to start after being inactive. | Wait a few seconds for the server to wake up and refresh the page if necessary.                         |